

Formula and Graph Documentation

Introduction

This documentation contains all formulas and graphical representations used in the “Expectation Decider” project. The project applies probability and statistical analysis techniques to student academic performance data. Different formulas are used to calculate probabilities, analyze relationships, and evaluate student performance patterns. Various graphs and visualizations are also included to represent the dataset in an understandable and meaningful way.

1. Probability Formula

The probability formula is used to calculate the likelihood of an event occurring. In this project, it is used to calculate the probability of students passing the exam, studying more than 10 hours, or maintaining high attendance.

Formula

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

Purpose

This formula measures the chance that a specific event will occur.

2. Empirical Probability Formula

Empirical probability is based on actual observed values from the dataset. It uses real student records for calculations.

Formula

$$P(E) = \frac{\text{Observed Outcomes}}{\text{Total Outcomes}}$$

Purpose

Used to measure real-world probability using actual student performance data.

3. Theoretical Probability Formula

Theoretical probability is calculated using assumptions where all outcomes are equally likely.

Formula

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Possible Outcomes}}$$

Purpose

Used to calculate ideal probability values.

4. Conditional Probability Formula

Conditional probability calculates the probability of one event occurring given another event has already occurred.

Formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Purpose

Used to calculate the probability of passing given high attendance or participation in group discussion.

5. Joint Probability Formula

Joint probability measures the probability of two events occurring together.

Formula

$$P(A \cap B) = \frac{\text{Students in Both Events}}{\text{Total Students}}$$

Purpose

Used to calculate the probability that students participated in group discussion and passed the exam simultaneously.

6. Marginal Probability Formula

Marginal probability measures the probability of a single event independently.

Formula

$$P(A) = \frac{\text{Total Pass}}{\text{Total Students}}$$

Purpose

Used to calculate overall student passing probability.

7. Binomial Distribution Formula

The binomial distribution formula calculates probabilities for repeated independent trials with two possible outcomes.

Formula

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Purpose

Used to calculate the probability of students passing among selected students.

8. Mean Formula

The mean formula calculates the expected number of successful outcomes.

Formula

$$\mu = np$$

Purpose

Used to calculate the expected number of students passing.

9. Variance Formula

Variance measures the spread of the probability distribution.

Formula

$$\sigma^2 = np(1-p)$$

Purpose

Used to measure variability in student performance.

10. Bayes' Theorem Formula

Bayes' theorem updates probability values using new evidence.

Formula

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Purpose

Used to calculate the probability that a student passed given high attendance.

11. Correlation Formula

The correlation formula measures the strength and direction of relationships between variables.

Formula

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Purpose

Used to analyze relationships between study hours, attendance, and previous test scores.

Graph Documentation

1. Histogram

Graph Used

Attendance Distribution Histogram

Purpose

A histogram is used to display the frequency distribution of attendance percentages among students.

Interpretation

The histogram shows how attendance values are distributed across different attendance intervals.

2. Bar Chart

Graph Used

Pass vs Fail Student Analysis

Purpose

A bar chart visually compares categorical values.

Interpretation

The graph compares the number of students who passed and failed the final examination.

3. Scatter Plot

Graph Used

Study Hours vs Previous Test Score

Purpose

A scatter plot shows the relationship between two numerical variables.

Interpretation

The graph indicates whether higher study hours improve previous test scores.

4. Venn Diagram

Graph Used

Students Studying More Than 10 Hours and Attendance Above 80%

Purpose

A Venn diagram visually represents relationships between sets.

Interpretation

The overlapping area represents students satisfying both conditions.

5. Heatmap

Graph Used

Correlation Heatmap

Purpose

A heatmap visually displays correlation values between variables.

Interpretation

The heatmap shows the strength of relationships between attendance, study hours, and scores.

6. Bayes Theorem Probability Graph

Graph Used

Probability Analysis Bar Graph

Purpose

Used to visualize probabilities used in Bayes' theorem calculations.

Interpretation

The graph compares overall passing probability, high attendance probability, and conditional probability of passing given high attendance.

7. Line Trend Graph

Graph Used

Probability Trend Line

Purpose

A trend line graph displays upward or downward movement in probability values.

Interpretation

The graph helps identify increasing or decreasing probability trends.

Conclusion

The formulas and graphical representations used in this project help analyze student academic performance using probability and statistics. Probability formulas provide mathematical calculations for student-related events, while graphs visually represent patterns, trends, distributions, and relationships within the dataset. Together, these techniques help understand how study hours, attendance, and group discussion participation influence examination success.